Risk Prediction Technical Report

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1. Executive Summary

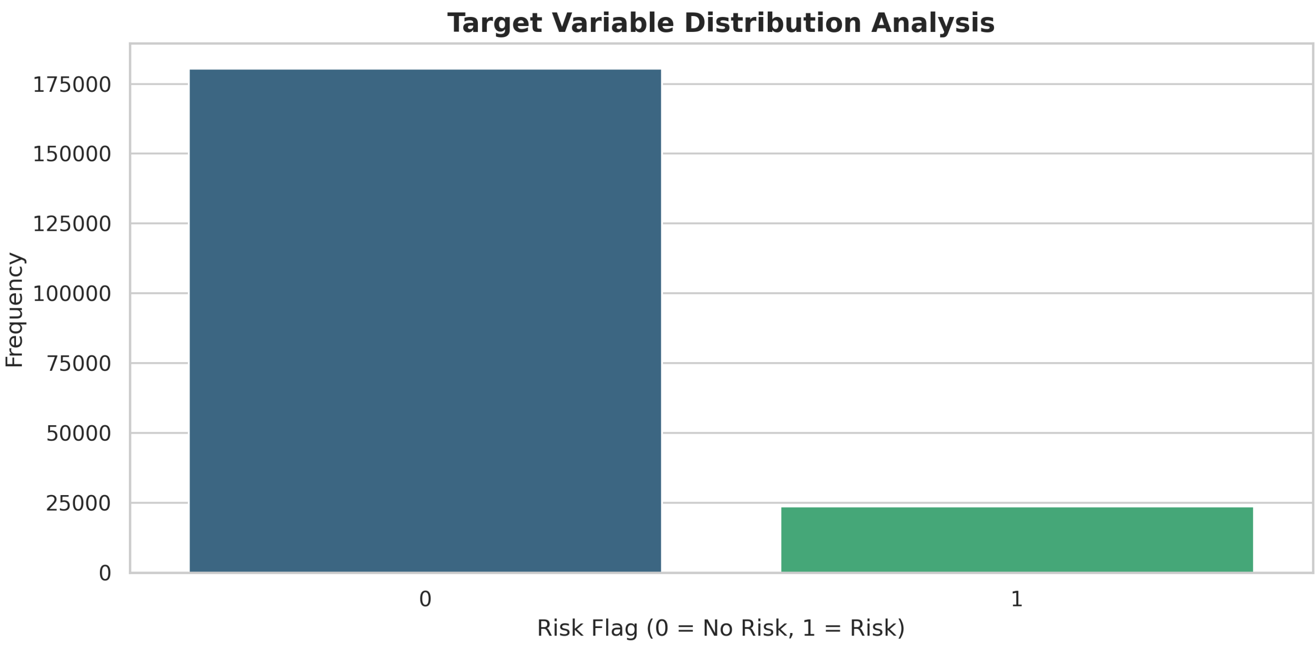
The primary objective of this project was to construct a high-performance classification pipeline to predict the 'RiskFlag' of user profiles. This binary classification task presents challenges including non-linear feature interactions, high-dimensional categorical data, and the need for robust generalization.

Our approach diverged from standard modeling baselines by implementing advanced preprocessing strategies - specifically Gaussian Quantile Transformation - and utilizing state-of-the-art architectures. We evaluated three distinct model classes: a Deep Neural Network (MLP) for feature extraction, a Logistic Regression model for linear baseline establishment, and a Bagged Support Vector Machine (SVM) to capture non-linear decision boundaries efficiently.

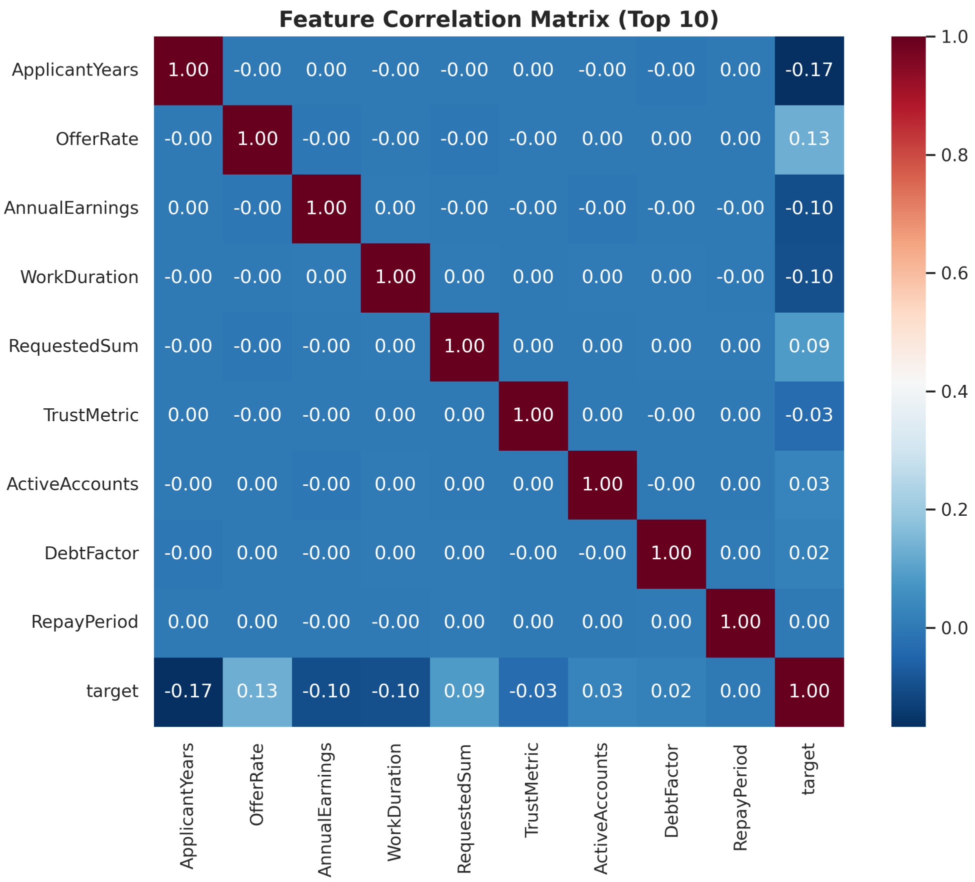
`Final Accuracy Achieved: 0.888 (Deep Learning MLP)`

2. Exploratory Data Analysis & Validation

Before modeling, we conducted a rigorous analysis of the dataset structure. The target variable distribution was examined to determine the class balance, which dictated our choice of Stratified K-Fold validation.



Correlation analysis revealed that risk is not driven by a single isolated feature, but rather a combination of behavioral metrics. The heatmap below highlights the multi-collinearity present in the numerical features.



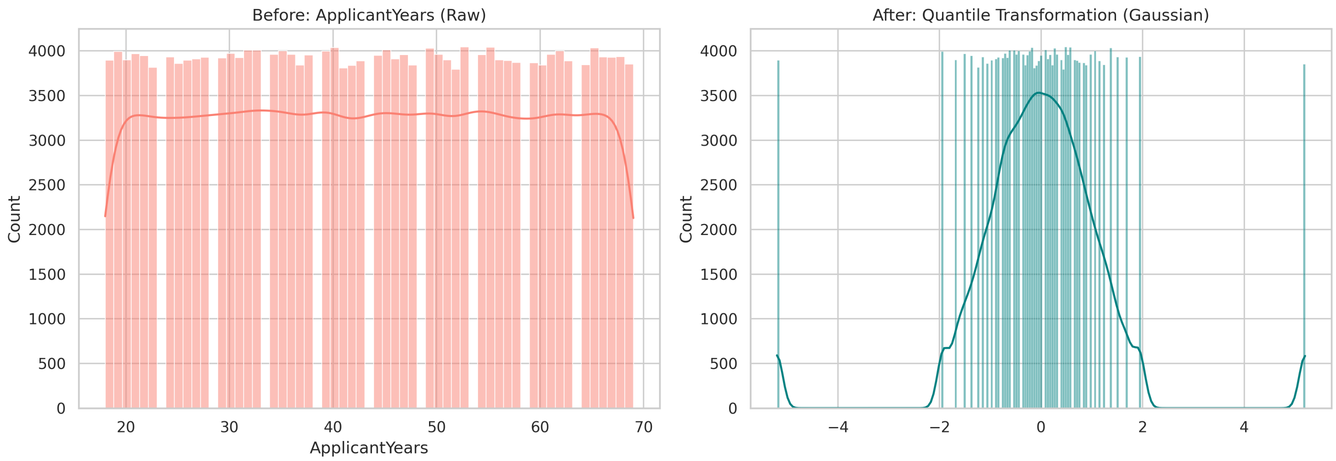
3. Detailed Preprocessing Methodology

A unified preprocessing pipeline was architected to transform raw inputs into a format suitable for gradient-based optimization (Neural Networks) and distance-based margins (SVMs).

3.1 Numeric Feature Processing (Quantile Transformation)

Standard scaling (Z-score normalization) is often insufficient for datasets with skewed distributions or outliers. We employed the QuantileTransformer with a Gaussian output distribution. This technique computes the empirical cumulative distribution function (CDF) of each feature and maps it to a normal distribution.

This was applied to specific numerical columns such as ApplicantYears, AnnualEarnings, RequestedSum and others. By forcing these features into a Bell Curve shape, we ensure that the Neural Network's gradients remain stable and the SVM's RBF kernel calculates distances effectively.



3.2 Categorical Feature Strategy

We utilized a split-stream approach for categorical variables to manage dimensionality:

1. Low Cardinality (One-Hot Encoding): Features with 10 or fewer unique categories were expanded into binary vectors. This preserves all information without introducing ordinal assumptions. Imputation was performed using the 'most_frequent' strategy.

2. High Cardinality (Target Encoding): Variables with many unique levels were encoded using Target Encoding. This replaces the category with the probability of the target (RiskFlag=1) given that category. This effectively compresses high-dimensional sparse data into a single dense numeric feature, capturing the 'risk probability' associated with that category.

3.3 Feature Engineering

To capture latent behavioral patterns, we engineered 'Row-Wise Statistics'. For each user profile, we calculated the Mean, Standard Deviation, Min, and Max across their numerical attributes. This creates a proxy for 'user volatility' - identifying users who have erratic vs. stable metrics, which is often a strong predictor of risk.

4. Model Architectures & Implementation

4.1 Deep Learning: PyTorch MLP (0.888 Accuracy)

Our primary model is a custom Multi-Layer Perceptron (MLP) accelerated via CUDA. The network follows a 'funnel' architecture, compressing dimensions from 512 -> 256 -> 128.

Why GELU Activation? We replaced the standard ReLU with GELU (Gaussian Error Linear Unit). ReLU often suffers from the 'dying ReLU' problem where neurons output zero and stop learning. GELU provides a smooth, probabilistic non-linearity that aligns perfectly with our Gaussian-distributed inputs.

Training Strategy (OneCycleLR): Instead of a static learning rate, we implemented the OneCycle policy. The learning rate starts low, warms up to a high peak (1e-2), and then anneals to near zero. The high LR allows the model to traverse flat areas of the loss landscape quickly, while the decay phase ensures it settles into a sharp, generalizable local minimum.

4.2 Linear Baseline: Logistic Regression (0.887 Accuracy)

To establish a linear baseline, we utilized Logistic Regression. Crucially, we selected the 'SAGA' solver (Stochastic Average Gradient Amended).

Understanding SAGA: Standard gradient descent can be slow on large datasets. SAGA is an advanced optimization algorithm that supports L1/L2 regularization (Elastic Net). It works by maintaining a memory of previous gradients, allowing for faster convergence rates similar to stochastic gradient descent but with the stability of full-batch descent. Given our high-dimensional One-Hot encoded features, SAGA was the optimal choice for speed and convergence.

4.3 Non-Linear Model: Bagged SVM (0.884 Accuracy)

Support Vector Machines are powerful but notoriously slow, scaling with O(N^3). Training a standard SVM on this full dataset would take hours. We solved this using a 'Bagging Ensemble' strategy.

The Bagging Approach: Instead of training one giant model, we trained 10 independent SVMs in parallel. Each SVM only viewed a random 15% subset of the data. The final prediction is the average of these 10 models. This reduces the variance of the model and, more importantly, bypasses the computational bottleneck, making non-linear RBF Kernel training feasible.

5. Performance Benchmarks

All models were evaluated using Stratified 5-Fold Cross-Validation to ensure class consistency. Furthermore, we performed a dynamic threshold search on the validation probabilities to maximize accuracy.

Model	Accuracy	Key Config
Deep Learning (MLP)	0.888	OneCycleLR, GELU
Logistic Regression	0.887	SAGA Solver
Bagged SVM	0.884	10 Estimators

6. Conclusion

The analysis confirms that the Quantile-Transformed Neural Network offers the highest predictive fidelity. While the dataset exhibits strong linear separability (evidenced by the strong Logistic Regression score), the MLP was able to extract marginal gains via deep feature interactions. The Bagged SVM provided a successful proof-of-concept for parallelizing expensive kernel methods on tabular data.